Enrolment Number										

Total No. of printed pages = 03

Winter, 2025

B.Tech 4th Semester Examination

Transportation Engineering – Theory

Course Code: BCE23401T

Full Marks – 50 **Time – 2 hours**

The figure in the margin indicates full marks for the questions.

SECTION 1

1. Choose the correct answer: **Marks: 10 x 1 =10**

I. What must be the height of crown of an earthen road of width 5.5 m passing through heavy rainfall area?
   a) 0.22
   b) 0.11
   c) 0.08
   d) 0.06

II. ________ terrain has ________ % cross slope.
   a) Plain, 10-25
   b) Rolling, 0-10
   c) Plain, 0-10
   d) Rolling, 25-60

III. What percentage camber must be provided for a CC road passing through low rainfall area?
   a) 3.0 %
   b) 2.5 %
   c) 2.0 %
   d) 1.7 %

IV. Full form of CRF is
   a) Central Rail Fund
   b) Central Road Finance
   c) Central Road Fund
   d) Central Road Facility

V. What is the ruling design speed on a state highway in a plain terrain?
   a) 70 kmph
   b) 60 kmph
   c) 100 kmph
   d) 80 kmph

VI. The value of longitudinal coefficient of friction ranges from ____________
   a) 0.10 to 0.30
   b) 0.25 to 0.30

c) 0.05 to 0.25
d) 0.35 to 0.40

VII. Bitumen is a by-product of ________
a) Wood
b) Petroleum
c) Kerosene
d) Coal

VIII. The layer which is constructed above embankment is called ________
a) Sub grade
b) Fill
c) Base
d) Sub base

IX. The radius of a horizontal circular curve is 100 m. The design speed is 50 kmph and the design coefficient of lateral friction is 0.15. Calculate the super-elevation required if full lateral friction is assumed to develop.
a) 1 in 21
b) 1 in 25
c) 1 in 30
d) 1 in 15

X. *In the CBR Test, the CBR ratio is usually determined for the penetration value of*
a) *2.5 mm*
b) 5 mm
c) a & b both
d) None of the above

SECTION 2

2. Answer any one question from this section **Marks: 8**

A) What are the basic requirements of an ideal alignment? Also discuss the surveys carried out before finalising the alignments of a highway.

B) Classify roads based on different parameters. Elaborate each category.

SECTION 3

3. Answer any one question from this section **Marks: 12**

A) Derive the equation for Stopping Sight Distance required in a highway. Also Calculate the Intermediate sight distance required on a highway if the design speed is 90 kmph. Assume necessary data.

B) What is a gradient? Mention its types. What will be the total width of National Highway on a horizontal curve of radius 450 m with a design speed of 65 kmph.

SECTION 4

4. Answer any one question from this section **Marks: 5**

A) What are the characteristics of traffic? Explain each point.

B) Discuss Traffic control devices, Traffic signals and Markings.

SECTION 5

Marks: 6

5. Answer any one question from this section

A) What are the tests carried out in bitumen? Explain ductility test in details.

B) How plate load test is carried out in soil for highway construction?

SECTION 6

Marks: 9

6. Answer any one question from this section

A) Write down the Construction Procedure of Bituminous Penetration Macadam Road.

B) A CBR test was conducted over soil sub grade and following results were obtained. Assume curve throughout convex following materials are required to be used over the soil sub grade.

i. Compact soil (CBR= 5.5%)

ii. Poorly graded gravel (CBR= 15%)

iii. Well graded gravel (CBR= 35%)

Design the pavement and find the thickness of the bituminous surface layer placed over the well graded gravel using CBR method if the wheel load is 4200 kg and tyre pressure is 7kg/cm^2.

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